Robust degree lower bounds for square-root polynomials with errors over finite prime fields

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Allowed standard prerequisites (editorial).

- Polynomial vanishing sets over finite fields and formal differentiation. Author Sections 1-3; original full bibliography retained

Theorem 1.1. *Let $p \equiv 1 \mod 4$. Then any polynomial that computes square roots in $\mathbb{F}_p$ on all but $e \leq \frac{p-7}{12}$ nonzero perfect squares in $\mathbb{F}_p$ must have degree at least $\frac{p-1}{3} - e$.*

Lemma 1.2. *Let $\mathbb{F}$ be a field of characteristic p. Let $f(X), A(X), B(X)$ be in $\mathbb{F}[X]$. Suppose*

$$f(X)^t = A(X) \cdot X^\ell + B(X) \tag{1}$$

where:

- $\deg(f(X)) \leq d < \frac{p}{t}$,
- $\deg(B(X)) \leq b < p$,
- $A(X) \neq 0$,
- $B(X) \neq 0$,

Then $d + b \geq \ell$.

In words, this says that if $dt < p$, the tth power of a polynomial of degree d cannot have d consecutive 0 coefficients.

Proof. Supose $d + b < \ell$. Observe that this implies that $f(X) \neq 0$.

Taking derivatives of both sides of (1), we get:

$$tf(X)^{t-1}f'(X) = C(X)X^{\ell-1} + B'(X), \tag{2}$$

for some $C(X) \in \mathbb{F}[X]$.

Computing $tf(X)^t f'(X)$ in two different ways using (1), (2), we get:

$$tA(X)f'(X)X^\ell + tf'(X)B(X) = f(X)C(X)X^{\ell-1} + f(X)B'(X).$$

Since $\deg(tf'(X)B(X)), \deg(f(X)B'(X)) < d+b \leq \ell-1$, by taking this equation mod $X^{\ell-1}$ we get:

$$tf'(X)B(X) = f(X)B'(X),$$

and since $f(X), B(X)$ are nonzero, we get that:

$$t\frac{f'(X)}{f(X)} = \frac{B'(X)}{B(X)}.$$

By the logarithmic derivative, we get $f(X)^t = \lambda B(X)$ for some nonzero λ, contradicting our assumption that $A(X) \neq 0$.

Thus $d + b \geq \ell$ as claimed. □

Theorem 1.1. *Let $p \equiv 1 \mod 4$. Suppose $f(X) \in \mathbb{F}_p[X]$ is a polynomial of degree d that computes square roots with error e.*

Then

$$d \geq \begin{cases} \frac{p-1}{3} - e & e \leq \frac{p-7}{12} \\ \frac{p-1}{2} - 3e - 1 & e > \frac{p-7}{12}. \end{cases}$$

Proof. We use the idea of the Berlekamp-Welch Reed-Solomon decoding algorithm [WB83].

Let $U \subseteq S$ be the set of $a \in S$ where $f(a)^2 \neq a$.

Let $E(X) \in \mathbb{F}_p[X]$ be the vanishing polynomial of U, given by:

$$E(X) = \prod_{u \in U} (X - u).$$

Note that E is a nonzero polynomial of degree at most e.

Then we have:

$$E(X)^2 \cdot f(X)^2 \equiv E(X)^2 \cdot X \mod (X^{(p-1)/2} - 1).$$

Let $A(X)$ be the polynomial of degree at most $2(e+d) - (p-1)/2$ such that:

$$E(X)^2 \cdot f(X)^2 - E(X)^2 \cdot X = A(X) \cdot (X^{(p-1)/2} - 1).$$

Let $g(X) = E(X) \cdot f(X)$, and $B(X) = E(X)^2 \cdot X - A(X)$.

Then

$$g(X)^2 = A(X) \cdot X^{(p-1)/2} + B(X).$$

We have:

- $\deg(g) \leq d + e$,
- $\deg(B) \leq \max(2e + 1, 2(e + d) - (p - 1)/2)$.
- $A(X) \neq 0$. Otherwise $E(X)^2 \cdot f(X)^2 = E(X)^2 \cdot X \implies f(X)^2 = X$, which is impossible.
- $B(X) \neq 0$. Otherwise $E(X)^2 \cdot X = A(X)$, and so $E(X)^2 f(X)^2 = E(X)^2 X^{(p+1)/2}$, which implies that $f(X)^2 = X^{(p+1)/2}$. This is only possible if $p \equiv 3 \mod 4$.

Plugging this into Lemma 1.2, we get:

- $$(d + e) + \left(2(d + e) - \frac{p - 1}{2}\right) \geq \frac{p - 1}{2},$$

  if $2d - (p - 1)/2 \geq 1$,

- $$(d+e)+(2e+1) \geq \frac{p-1}{2},$$

  if $2d-(p-1)/2 \leq 0$.

This tells us that either:

$$d \geq \frac{p-1}{3} - e,$$

or:

$$d \geq \frac{p-1}{2} - 3e - 1,$$

and thus:

$$d \geq \min\left(\frac{p-1}{3} - e, \frac{p-1}{2} - 3e - 1\right),$$

which gives us the desired claim. □

Let p be an odd prime, and let $\mathbb{F}_p$ be the finite field with p elements.

We say $f(X) \in \mathbb{F}_p[X]$ *computes square roots* in $\mathbb{F}_p$, if for all nonzero perfect squares $a \in \mathbb{F}_p$, we have:

$$f(a)^2 = a.$$

In other words, for each nonzero perfect square $a \in \mathbb{F}_p$, $f(a)$ is one of its two square roots.

Logarithmic derivatives

The key fact about logarithmic derivatives that we are using is that if $f(X), B(X) \in \mathbb{F}_p[X]$, $t \cdot \deg(f), \deg(B) < p$, and:

$$t\frac{f'(X)}{f(X)} = \frac{B'(X)}{B(X)},$$

then $f(X)^t = \lambda \cdot B(X)$ for some constant $\lambda \in \mathbb{F}_p$.

We recap a quick proof. The hypothesis implies that $\left(\frac{f(X)^t}{B(X)}\right)' = 0$, and thus $\frac{f(X)^t}{B(X)}$ must be a rational function in X^p. To see the last deduction, note that $\left(f(X)^t \cdot B(X)^{p-1}\right)' = \left(\frac{f(X)^t}{B(X)} \cdot B(X)^p\right)' = \left(\frac{f(X)^t}{B(X)}\right)' \cdot B(X)^p + \frac{f(X)^t}{B(X)} \cdot p \cdot B(X)^{p-1} \cdot B'(X) = 0 + 0 = 0$. Thus the polynomial $f(X)^t \cdot B(X)^{p-1}$ is a polynomial in X^p, which implies that $\frac{f(X)^t}{B(X)} = \frac{f(X)^t \cdot B(X)^{p-1}}{B(X)^p} = \frac{f(X)^t \cdot B(X)^{p-1}}{B(X^p)}$ is a rational function in X^p. Once we know that $\frac{f(X)^t}{B(X)}$ is a rational function in X^p, our assumption about the degrees implies the result.

Let p be a prime that is 1 mod 4. Let S be the set of nonzero perfect squares in $\mathbb{F}_p$.

We say a polynomial $f(X) \in \mathbb{F}_p[X]$ *computes square roots with error* e if:

$$\left|\{a \in S \mid f(a)^2 \neq a\}\right| \leq e.$$

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